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Tooling Sustainability Program — First 5 Funded Report

The Open Source Committee (OSC) has dedicated significant time to ensure that proper review is taking place with regards to funding various Cardano tools. The **First 5 Funded Report** documents progress on the first five tools funded under the program, covering milestone delivery, adoption evidence, and compliance status against the program's funding rules. To date of publication, **6 milestones** have tracked across the five funded tools, totaling **~431,730ADA / $95,250** disbursed against completed deliverables.

Foreword by the Open Source Committee

The Tooling Sustainability Program funds open-source tools that Cardano developers, operators and governance bodies rely on but that have no stable source of funding. This report covers its first five funded tools, spanning governance, off-chain development, cross-language interoperability, AI-assisted development and contract testing.

Two awards show the program's range:

- **Cicerium**, the first award, was developed to move governance-document review from off-chain channels to on-chain via a Hydra-executed process anchored on L1.
- **Cardano Transaction Library (CTL)**, a dependency of production projects such as Indigo and Liqwid, was funded to stay compatible through the van Rossem hard fork, the kind of upkeep the program is built for.

Since the first award, the OSC has progressively tightened its review, giving more weight to active, demonstrated use. Every project is paid only against milestones with agreed acceptance criteria, reports progress each quarter, and is re-evaluated each budget cycle with no automatic renewal. LambdaBuffers' funding, for example, is tied to its move from an internal dependency to a governed public good.

1. Cicerium

Voting tool: Originally designed as a Hydra-based feedback and peer review system for the 2026 Cardano Budget Process, revised for other decentralized governance decisions.
Band & Award: Band 1 — $24,000 USD, paid in ADA (~91k)
Program Start: 2026-02-02
Repository: https://github.com/iamhammadaali/cicerium

Cicerium was the first tool funded under the Tooling Sustainability Program and served as a test case for its process. It was approved quickly on the assumption it would be used in the 2026 Budget Process. Following review and contract completion, the OSC determined that future tooling projects would undergo a more detailed review targeting explicit delivery and improvement, with a higher emphasis on proven, active adoption.

Milestone 1 — Requirements alignment with the Civics Committee: Meet with the Civics Committee and its working groups, review current tool capabilities, and define the delta of work needed to meet the committee's needs.

Milestone 2 — Pre-production deployment: Deploy to a pre-production environment, run usability testing with the Civics Committee and SMEs, deliver usability reports, and implement resulting updates.

Milestone 3 — Production deployment**:** Deploy to production and monitor initial usage through the Cardano Constitution Amendment Process.

Adoption evidence:

Funded on the basis of a committed use case rather than prior adoption. Delivered per the milestone-based contract moving governance-document review from off-chain channels to a Hydra-executed, L1-anchored process with reviewer accountability.

Compliance status: Self-attestation: Yes, Dec. 03, 2025 · License: MIT

2. [Plutonomicon/CTL]

CTL has been a core off-chain library since the Alonzo era and is a foundational dependency for a number of production projects, but had no secured funding source to remain compatible with the 2026 hard fork. The award is scoped narrowly to hard-fork readiness rather than general maintenance, with delivery via public PRs to the CTL repository, a two-month term, and third-party delivery assurance by No Witness Labs.

Band & Award: Band 1 — $30,000 USD, paid in ADA (~170k)
Program Start: 2026-07-28
Repository: github.com/Plutonomicon/cardano-transaction-lib

Milestone 1 — Basic Compatibility (month 1): Update CTL's runtime dependencies (Nix flake, Spago) to Protocol Version 11-compatible releases of Cardano Node 11.0.1, Ogmios, and Kupo; verify the cardano-testnet-based local cluster and testing utilities work against the updated node; run the full unit, integration, and end-to-end suites, fixing or documenting any failures, and confirm the Blockfrost backend does not regress.

Milestone 2 — Full Van Rossem Support and Release Readiness (month 2): Audit and update the balancer and transaction-building pipeline for the revised Plutus V1/V2 reference-input rules, with dedicated tests; review and close gaps in governance transaction support; update documentation and tutorials so downstream projects can migrate to the compatible release; deliver a written compatibility report covering changes, testing summary, and any deferred work.

Adoption evidence: CTL is integrated in production by projects: Indigo, Genius Yield, Liqwid, Cardano Racers, Noble Steed Games, and SingularityNET, with earlier adopters including DCSpark and TxPipe. The main repository has 50+ forks and 90+ stars, with maintained downstream forks by Liqwid Labs and Fourier Labs, and a history of external contributions and issue reports from Fourier Labs, BinarApps, Tontine Trust, Platonic Systems, and Indigo.

Compliance status: Self-attestation: Yes, Oct. 27, 2025 · License: MIT

3. LambdaBuffers

LambdaBuffers was accepted as a strategic investment in cross-language interoperability: it closes the "serialization gap" between on-chain Plutus data and off-chain services, a common source of silent, high-risk bugs in dApp architectures. The OSC noted low development velocity and high maintainer concentration at the time of acceptance, and funding is tied to the project's transition from an internal dependency to a governed public good, with mandates to resolve compliance gaps, adopt the Intersect Governance Policy, and use public channels for project decisions.

Band & Award: Band 2 — $15,000 USD, paid in ADA (~80k) (accepted at Band 3, upgraded to Band 2 by OSC vote)
Program Start: 2026-07-28
Repository: github.com/mlabs-haskell/lambda-buffers

Milestone 1 — Dependency Updates (months 1–2): Update the LambdaBuffers Nix dependency set to current compatible versions on the staging branch, repair and validate the Nix development shell so it launches without errors and exposes all dependencies needed to build and develop the project, and record the changes in CHANGELOG.md.

Milestone 2 — Van Rossem Compatibility Updates (month 3): Update all LambdaBuffers backends, and any MLabs-owned external dependencies, for Van Rossem support; add and validate tests for new functionality; and publish a new tagged release (version above 2.0.0) with a passing CI run and updated CHANGELOG.md.

Adoption evidence: Stable v1.0.0 release with multi-language code generation and prior Catalyst-funded development. At acceptance the OSC recognised the project's technical maturity while noting that external adoption was not yet demonstrated; broadening use beyond the maintaining organisation is a stated objective of the funding.

Compliance status: Self-attestation: Yes, Nov. 27, 2025 · License: Apache 2.0

4. [MeshAI]

MeshAI addresses a gap specific to AI-assisted development on Cardano: general-purpose models generate incorrect or outdated code for the eUTXO model, and full SDK documentation exceeds typical LLM context windows. It was funded as the first Cardano-specific AI developer tooling, building on MeshJS's established user base. The OSC noted that MCP standards and agent frameworks are evolving quickly, making the program's annual re-evaluation particularly relevant for this project.

Band & Award: Band 2 — Award amount to be set once a SoW is agreed
Program Start: Not yet started (pending SoW)
Repository: github.com/MeshJS/Mesh-AI

Status - approved, awaiting Statement of Work: MeshAI was approved by the OSC and is still eligible for funding. The OSC remains willing to fund this tooling, but the Statement of Work process has stalled. No workable, fundable SoW has been received, so no milestones or award amount have been set and no funds have been disbursed. The next step sits with the MeshJS team, and MeshAI's final status will be recorded in the program's future report.

Milestones - To be defined in the agreed Statement of Work.

Adoption evidence: Live production deployments at meshjs.dev/ai (documentation platform with AI chat), meshjs.dev/ai/chatbot, and meshjs.dev/ai/mcp, compatible with Claude Code, Cursor, Codex, and a broad set of AI agents. Because MCP servers and agent skills run locally in developers' tools, direct usage telemetry is not available; adoption is currently evidenced through developer feedback and community demand for AI-compatible documentation.

Compliance status: Self-attestation: Yes, Jan. 31, 2026 · License: Apache 2.0

5. Cardano-node-emulator

The emulator is hosted under the IntersectMBO organisation and maintained by Tweag. It was funded as security-focused testing infrastructure: it exposes the ledger's validation logic in a local environment, reducing the dependency on public testnets for contract testing and audit tooling, and also provides utilities for building the contracts under test.

Band & Award: Band 1 — $50,000 USD (OSC Cost: ~238k ADA)
Program Start: 2026-09-04
Repository: github.com/IntersectMBO/cardano-node-emulator

Milestone 1 —

- Establish precise roadmaps for the update and maintenance of CNE, for both the present engagement and the assigned community maintainer.
- Refine this Statement of Work into manageable subtasks and precise planning (the roadmaps do not replace this SoW).

Milestone 2 —

- Update the full CNE project to the most recent dependencies by updating cardano-api, which in turn pulls new versions for the other dependencies.
- Update each of the five projects and prepare a release.

Milestone 3 —

- Address leftover issues across plutus-script-utils, plutus-ledger, freer-extra, cardano-node-emulator and cardano-node-socket-emulator (dependency clean-up, efficiency, repository restructuring, genesis configuration, interface clean-up, and assessing the socket emulator).
- Submit the resulting work for release on CHAP.

Milestone 4 —

- Engage the community about CNE via open-source channels, regular Discord updates and other means.
- Write and publish a blog post on the Tweag blog.

Adoption evidence: Wider use by contract developers as a local testing environment is a stated objective of the funding.

Compliance status: Self-attestation: Yes, July 6, 2026 · License: Apache 2.0

Conclusion

Overall, milestone-gated funding is working as intended across the first five tools. Funds have been released against completed deliverables, and none are committed until a workable scope is agreed.

The next reporting period will focus on delivery against the remaining milestones and on demonstrated use of each funded tool, which the OSC now weighs more heavily in its reviews.